

CSCE 650 VIRTUAL REALITY · MODULE 08 · LECTURE NOTES

Locomotion

Moving through a space larger than the room.

MODULE OUTCOMES

- Compare locomotion schemes on comfort and presence.
- Explain why teleport is comfortable and what it costs.
- Explain redirected walking and its space requirements.
- Apply the comfort techniques that make smooth locomotion viable.
- Choose a locomotion model from the experience being built.

1 The constraint

REAL WALKING IS THE BEST OPTION AND DOES NOT SCALE

Walking in the physical space is perfectly comfortable. The vestibular and visual signals agree exactly, because both are reporting a real motion that is genuinely occurring. There is no conflict to resolve (Module 07).

It also produces the strongest sense of presence of any locomotion method, by a considerable margin — the body's own proprioceptive and effort cues are present and consistent.

And a typical play space is about 2 metres square. Many users have less; many are seated; some are on an aeroplane.

So every technique in this module is a way of covering more distance than the room physically permits, and every one of them is worse than real walking in at least one respect. There is no solution here, only a set of trades.

Scheme	Comfort	Presence	Space	Notes
Real walking	Perfect	Highest	As much as the virtual space	The benchmark everything else is measured against.
Teleport	Excellent	Low	Any, including seated	§2. The safe default.
Dash / blink	Very good	Low to moderate	Any	A very fast movement with a fade — between teleport and smooth, and a reasonable compromise.

Scheme	Comfort	Presence	Space	Notes
Smooth (continuous)	Poor without mitigation	High	Any	§3. What experienced users ask for.
Arm swinging	Good	Moderate	Any	Swing the controllers to walk. The physical effort appears to help, and it is tiring.
Redirected walking	Excellent	Highest	~6×6 m minimum	§4. Wonderful and rarely applicable.
Vehicle / cockpit	Very good	High within the fiction	Seated	The cockpit supplies a stable reference frame (Module 07), which is why driving and flying worked in VR years before walking did.

The comfort and presence columns are in tension throughout, and that tension is the entire design problem of this module. No scheme is top of both columns and fits in an ordinary room.

2 Teleport

TELEPORT WORKS BECAUSE THERE IS NO MOTION TO CONFLICT WITH

The user aims, confirms, and is instantly somewhere else. No visual motion occurs at all, so novection is induced, so there is nothing for the vestibular system to contradict.

It is comfortable for essentially everyone, including the substantial group of people who cannot tolerate any continuous scheme. For an application that needs to be usable by an unselected audience, this is decisive.

The cost is presence and spatial awareness. Moving discontinuously through a space is nothing like being in it, and users lose track of where they are relative to where they were — the cognitive map that real walking builds for free does not form.

A brief fade or blink mitigates the disorientation without reintroducing motion. It signals that a transition has occurred rather than animating the transit, which is the important distinction.

- **Show the destination clearly before committing.** A projected arc, a marker on the ground, and an indication of which way the user will be facing. Uncertainty about where you will arrive is the main usability failure of bad teleport.
- **Indicate validity during the aim.** The arc should change colour or visibly break when the target is unreachable, so the user learns the rules without having to fail.
- **Let the user choose their facing direction during the aim** — typically by the thumbstick's direction. Arriving and then having to turn is both an extra interaction and a comfort problem, since turning is the provocative motion. **This is the refinement that distinguishes a good teleport implementation from a basic one.**

- **Keep the transition brief** — a fade of around 100 ms. A long fade is an animation, and animations of motion are what we are avoiding.
- **Preserve the spatial relationship.** Teleporting should not silently change the user's height, their orientation relative to the room, or their position within the play space, all of which are disorienting in ways users cannot diagnose.

3 Smooth locomotion

HIGH PRESENCE, HIGH RISK

Continuous movement feels like being in a place in a way teleport simply does not. The space is traversed rather than sampled, distances are felt, and the cognitive map forms. Experienced VR users strongly prefer it and frequently ask for it.

And it produces vection continuously, which is precisely the mechanism Module 07 identified. The user is visually moving and physically still, for as long as they hold the stick.

It is the single largest cause of sickness in shipped VR applications. Not latency, not frame drops — locomotion.

So it must be offered as an option and never imposed as the only mode, and it must be built with the mitigations below from the start rather than having them retrofitted. An application whose only locomotion is unmitigated smooth movement has excluded a large fraction of its potential users by construction.

Technique	Effect	Notes
Constant velocity, no acceleration	Large	The most important single measure (Module 07). Reach full speed immediately and stop immediately; a gentle ramp is worse than an abrupt one.
Dynamic vignette during movement	Large	Darken and narrow the periphery while moving, restore it when stationary. Attacks vection at its strongest point. Make the strength adjustable.
Snap turning	Large	See below.
Lower movement speed	Moderate	Less visual flow per second. Too slow becomes frustrating and users turn it up, so this trades against itself.
A cockpit or helmet frame	Moderate	A stable reference frame (Module 07).
Head-relative rather than controller-relative direction	Modest	Moving where you look is more predictable than moving where the controller points, and predictability reduces conflict. Offer both; preferences differ strongly.

SNAP TURNING IS NOT A COMPROMISE

Visually rotating the world while the head is stationary is the most provocative item in Module 07's table. The semicircular canals are exquisitely sensitive to rotation, and a smooth stick-driven turn produces a large, sustained, entirely uncontradicted visual rotation signal.

Snap turning jumps the view by a fixed increment — 30 or 45 degrees — usually with a brief fade. **No continuous rotation ever occurs**, so the canals have nothing to object to.

Users adapt within a minute and generally stop noticing it. The common objection — that it is unnatural — turns out to matter much less in practice than it does in discussion.

And physical turning should always remain available. Many users will simply rotate their body, which costs nothing and is perfectly comfortable. **Designs that assume a forward-facing user** — by placing important content only ahead, or by binding a cable awkwardly — discard the best turning method available.

4 Redirected walking

VISION DOMINATES PROPRIOCEPTION, AND THAT CAN BE EXPLOITED

When visual and proprioceptive signals disagree slightly, vision wins and the discrepancy goes unnoticed. This is a robust finding and it is the basis of the entire technique.

So rotate the virtual world slowly while the user walks what they believe is a straight line. They will in fact walk a curve in the physical room, continuously correcting without awareness, while experiencing a straight path through the virtual space.

The detection thresholds have been measured: roughly 10 degrees per second of added rotation while the user is already turning, and a curvature radius of about 22 metres while walking straight, both go unnoticed by most people.

The catch is the space requirement. A 22-metre curvature radius implies a physical room of roughly 6 by 6 metres at absolute minimum to keep a user walking indefinitely — and realistically more. **This is why redirected walking, which is genuinely excellent, appears almost exclusively in research labs and location-based venues.**

- **Rotation gain** amplifies the user's own turns: a 90-degree physical turn produces a 100-degree virtual one. Subtle, effective, and it accumulates usable reorientation over a session without any single turn being detectable.
- **Curvature gain** bends the virtual path so the physical path curves. The most powerful technique and the one with the largest space requirement.
- **Change blindness** moves geometry while the user is not looking at it — a doorway relocated to the other end of a room while the user's back is turned. **It works remarkably well**, because people do not monitor the parts of an environment they are not attending to.

- **Impossible spaces** let rooms overlap in physical space, connected by corridors whose virtual length exceeds their physical one. A user walks between two rooms that could not both fit. **This is the most practical of these techniques for ordinary room sizes and is underused.**
- **Resetting** handles the failure case: when the user approaches a physical wall despite everything, intervene — distract them into turning around, or present an in-fiction reason to reorient. Inelegant, and necessary.

CHOOSE THE LOCOMOTION MODEL BEFORE ANYTHING ELSE

Locomotion determines what the experience can be. A teleport game and a smooth-locomotion game are different games that happen to share assets — not one game with a settings toggle.

It constrains level design directly. Teleport requires valid destinations, clear sightlines to them, and spaces that read correctly when sampled discontinuously. Smooth locomotion requires continuously navigable geometry and tolerable sightline changes. These produce different level geometry.

It constrains the fiction. Why can this character teleport? Many designs answer by making the player a machine, a ghost, or a pilot — and the cockpit answer solves the comfort problem at the same time, which is not a coincidence.

Retrofitting locomotion is a redesign, not an adjustment. This is why Module 07 said to decide it first, and it is the single most consequential early decision in a VR project.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 10 (free)	Locomotion and its relationship to sickness, with the research framing.
Steinicke et al. — Estimation of Detection Thresholds for Redirected Walking (free)	Where §4's threshold numbers come from. The experimental design is worth reading on its own.
Suma et al. — Impossible Spaces and Change Blindness Redirection (free)	The two most practical redirection techniques for small rooms.
Meta — locomotion design guidelines (free)	The practical comfort rules for each scheme, with vendor testing behind them.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement teleport with an arc, a validity indicator, and facing-direction selection during the aim. Test on a novice.
- 2 Implement smooth locomotion with and without a dynamic vignette and compare comfort scores across six people.
- 3 Implement acceleration ramps and instant velocity changes, and compare. The result is counterintuitive.
- 4 Implement snap turning and smooth turning. Measure how long it takes users to stop noticing snap turning.
- 5 Implement rotation gain and find your own detection threshold by increasing it until you notice.
- 6 Implement an impossible space: two rooms that overlap physically, connected by a corridor. Measure whether anyone notices.
- 7 Implement change-blindness redirection by moving a doorway while the user is not looking at it.
- 8 Measure how far a user can actually walk before hitting a wall, with and without redirection, in your available space.
- 9 Write the locomotion section of a design document for an application of your choosing, justifying the choice from this module and Module 07.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is real walking the benchmark, and why is it impractical?
- 2 Compare seven locomotion schemes on comfort, presence, and space.
- 3 Why is teleport comfortable, and what does it cost?
- 4 Give five design rules for good teleport, and say which is most often missed.
- 5 Why is smooth locomotion both preferred and dangerous?
- 6 Name the three mitigations that make smooth locomotion viable.
- 7 Why is snap turning not a compromise?
- 8 How does redirected walking work, what are the thresholds, and what space does it need?
- 9 Name four redirection techniques and say which suits a small room.
- 10 Why must locomotion be chosen first?